

The Best Chips For MOBILE DEVICES



Vertica Asthana
is a technical
journalist at EY

A system-on-a-chip (SoC) integrates digital, analogue and mixed signals, as well as RF functions into a single unit. It consumes less power, which makes it ideal for smartphones.

Lower the power consumed by a chipset, higher is its efficiency. To ensure this, chipset technology has now reached 10nm scale, where key SoC manufacturers such as Qualcomm, Spreadtrum, Mediatek, Apple, Huawei and Intel are competing to outdo each other in chip performance and capture markets worldwide.

However, 10nm is not the final limit. "It is predicted that 10nm technology will be used till the end of year 2017. Year 2019 will experience a new 7nm technology, while around 2022 5nm technology would be in the market," says Jag Bolaria, principal analyst for embedded and servers at Linley Group.

Features embedded in SoCs

An SoC is equivalent to the motherboard of a computer system that comprises a GPU, CPU

and memory on a single chip. It is a successful integration of multiple domains such as device, design and application. As far as the device is concerned, several transistors are embedded into one single chip. This allows the chip to perform a wide range of functions through a single unit (chip).

As the increasing number of transistors on the chip push its applications to a broader range, chip design is becoming complicated and challenging. The challenge could be related to the device, design or execution.

SoCs comprise internal components such as microcontrollers/microprocessors, memory blocks, oscillators and phase-locked loops (PLLs) for timing, counter-timers/real-time timers, external interfaces such as USB, Ethernet and FireWire, analogue interfaces, voltage regulators and power management circuits.

Evolution in SoC technology

Various manufacturers use different configurations inside their SoCs depending on the type of technology used. For instance, Nvidia Tegra 3 SoC, built using 40nm technology, has only a GPU, memory controller and video out streams. On the other hand, Qualcomm Snapdragon S4, fabricated using 28nm technology, has LTE, GPE and WiFi all integrated in a single chip.

Newer chipsets fabricated on 10nm technology act as a high-end platform for highly efficient systems. At the same time, there are a few manufacturers struggling to roll over to mid-level and low-level smartphone segments with fabrication above 10nm.

TABLE I
CHIPSETS FOR PREMIUM SMARTPHONES

System-on-chip	Manufacturer	Technology	CPU core (no.)	Memory type	Geekbench 4.1 64-bit single-core score	Geekbench 4.1 64-bit multi-core score
Apple A11 Bionic	Apple	10nm	6	LPDDR4	4061	9959
Samsung Exynos 8895	Samsung	10nm FinFET	8	LPDDR4	2015	6711
Snapdragon 835	Qualcomm	10nm	8	LPDDR4	2065	6436
HiSilicon Kirin 960	Huawei	16nm	8	LPDDR4	1859	6356
Mediatek MT 6799	Mediatek	10nm	10	LPDDR4	1579	3668
Intel Atom Z3580	Intel	22nm	4	LPDDR3	1046	2515